

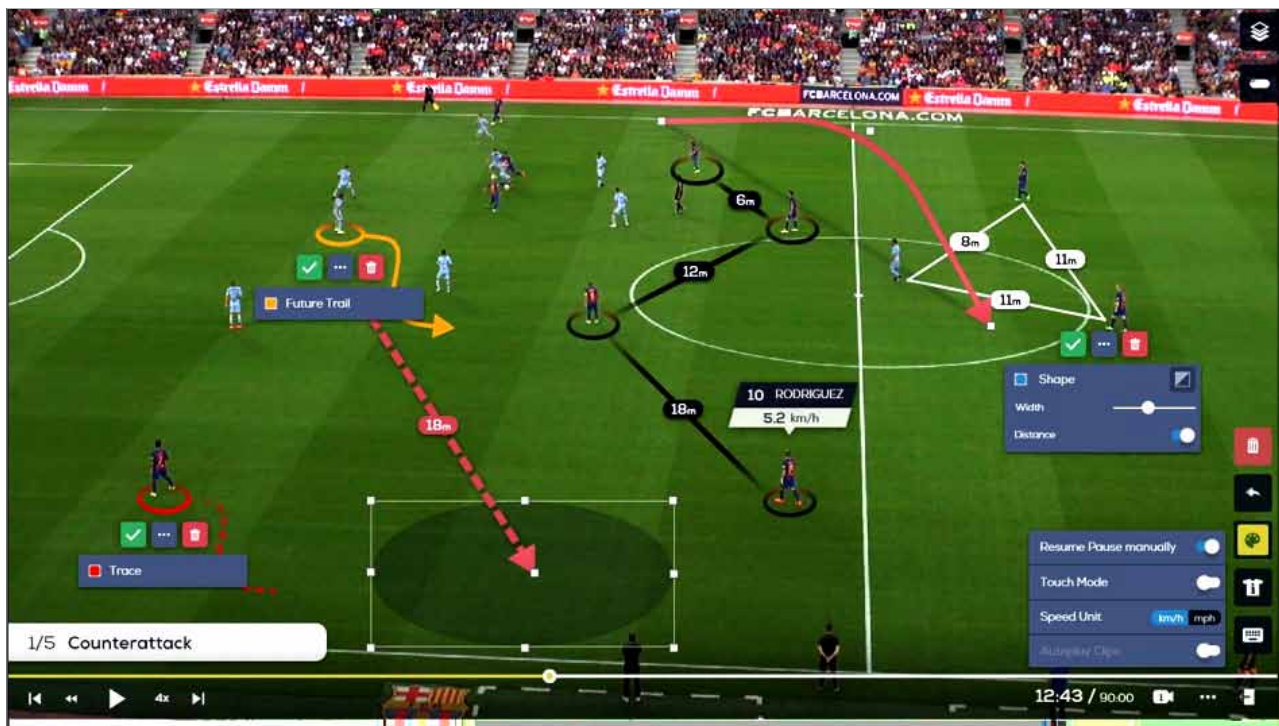


Figure 1: Sports data analysis with movement and video data

Since neural networks used in DL are trained rather than programmed, applications following this approach often require less fine-tuning and expert analysis. The availability of a humongous amount of video data in today's system supports this cause. While computer vision algorithms tend to be more domain-specific, DL, on the other hand, provides superior flexibility because CNN (convolutional neural networks, a popular model in deep learning) models can be retrained using a custom data set for any use case without requiring manual engineering of features.

Deep learning has found widespread applications in healthcare for diagnosing diseases from radiological images, in autonomous vehicles for object detection and autonomous driving, in finance for fraud detections, as well as in retail and ecommerce, manufacturing and industry, security and surveillance for facial recognition, and more.

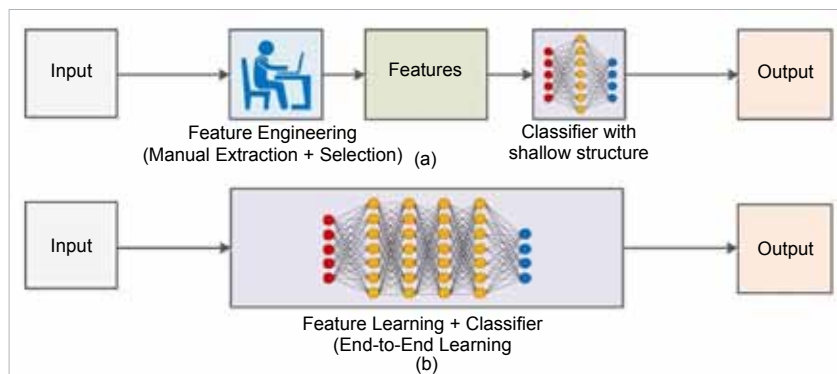


Figure 2: (a) Traditional computer vision workflow; (b) Deep learning workflow

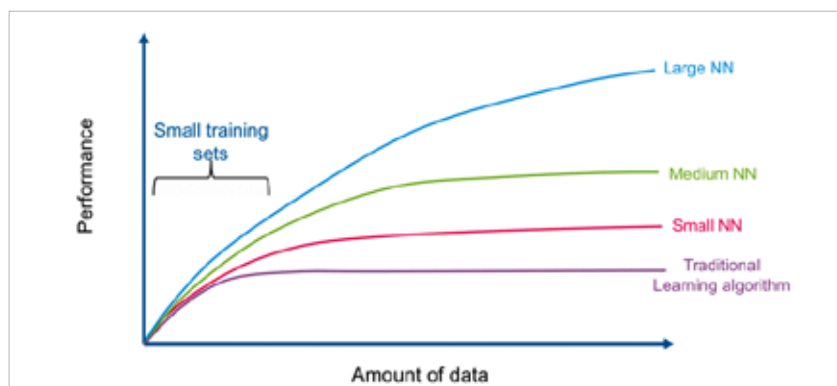


Figure 3: Comparing the performance of deep learning-based algorithms and traditional algorithms